

Individual Web Element Contribution

Quick Reference Guide for Translink Reporting
System

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1. CREATE - Add Records via Web Interface (2 pages)

createStop.html and createRouteStop.html and createScript.js

Validation

- This is a selection of my validation for both of my create pages. The first stops null data from being submitted and the second disallows the use of special characters in the names to ensure consistency and accuracy for stop names.

//createScript.js - Line 101 - 103, 109 - 111

```
if (!stopName) {  
    return "Stop name is required.";  
}  
  
if (!/^[a-zA-Z0-9\s]+$/.test(stopName)) {  
    return "Stop name must only contain letters,  
    numbers, and spaces.";  
}
```

Maintain Data Integrity

- This prevents duplicate route-stop records being created – there is also a filter (not shown here) to ensure the stop type is correct for the route

//createRouteStop.html - Line 298 + 301

```
AND NOT EXISTS(  
    SELECT 1  
    FROM tblRouteStop rs  
    WHERE rs.routeId = ${routeMap[routeText]}  
    AND rs.stopId = s.stopId)
```

2. UPDATE - Modify Existing Stop Records

updateStop.html + createScript.js (not for code examples)

Stop ID	Stop Name	Stop Type	Action
1	Grand Central Station	Bus & Train Station	<button>Edit</button>
2	City Hospital Train Station	Train Station	<button>Edit</button>
3	Botanic Train Station	Train Station	<button>Edit</button>
4	Tamnamore Park and Ride	Bus Stop	<button>Edit</button>

Stop name must only contain letters, numbers, and spaces.

Stop record updated successfully.

Choose a stop from the table before updating.

Edit Selected Record

Editing Stop ID 8

Stop Name

Stop Type

Train Station

-- select an option --

Bus Stop

Train Station

Bus & Train Station

Buscentre

Validation

UPDATE operations reuse existing validation rules and apply changes using a primary key to ensure only the intended record is modified.

//UPDATE is tied to a specific record ID - **Line 208**

```
selectedStop = Number(stop.stopId);
```

//Reuses the same client-side validation rules as CREATE before applying updates. - **Line 383**

```
const validationMessage = validateStop(stop);
```

Maintain Data Integrity – Controlled Updates

Prevents updates that would conflict with existing route–stop relationships

//identifies if stop is part of a train route - data integrity - **Line 286**

```
const hasTrainRoutes = routeNames.some(name =>  
name.includes("line"));
```

//only update currently selected stop - **Line 393**

```
WHERE stopId = ${selectedStop};
```

3. DELETE - Remove Unused Feature Records

deleteFeature.html

Delete a Feature

The table below shows all Deletable Features. Click delete to remove a record.

All Unused Features are listed below.

Feature ID	Feature Name	Feature Category	Action
16	Adaptive Haptic Audio Shelter	Stop	Delete

Deletion cancelled.

Feature record deleted successfully.

This page says

This action cannot be undone. Are you sure you want to delete "Adaptive Haptic Audio Shelter" feature?

OK

Cancel

Delete Confirmation

```
//User confirmation is required before deletion
- Line 211 - 216
const shouldDelete = confirm(`Delete Feature
${row.featureId}?`);
if(!shouldDelete) {
    showMessage("Deletion cancelled.", "error");
    return;
}
```

Maintain Data Integrity – Controlled Delete Options

Prevents deletes that would risk creating invalid foreign keys (in tblVehicleFeature or tblStopFeature)

```
// Enforce referential integrity by preventing
deletion of in-use records - Line 166
WHERE sf.featureId IS NULL AND vf.featureId IS NULL

// Delete is only permitted for records with no
existing associations - Line 218
const deleteSQL = `DELETE FROM tblFeature
WHERE featureId = ${row.featureId}`;
```

4. REPORT – Routes Serving Hospital Stops

Rosa_Hospitals.html



Add Records

Update Records

Delete Records

Complaint Reports

Feature Reports

Routes And Stops Reports

Vulnerable Passenger Reports

How many transport routes include at least one stop serving a hospital?

The table below shows Translink routes that serve one or more hospital locations, including route start and end points and a list of hospital stops within the routes

Hospital stops were identified based on stop features containing the term 'hospital' within our database.

Number of Hospital Stops: 7

Routes with Hospital Stops

Number of Routes with Hospital Stops: 26

Route Name	Start Location	End Location	Hospital Stops	# Stops
G1	Belfast	Dundonald	Royal Hospitals Falls Road, Ulster Hospital	2
G1	Dundonald	Belfast	Royal Hospitals Falls Road, Ulster Hospital	2
10	Belfast	Portaferry	Ulster Hospital	1
10	Portaferry	Belfast	Ulster Hospital	1
1A	Derry-Londonderry City Centre	Wheatfield	Altnagelvin Hospital	1
1A	Wheatfield	Derry-Londonderry City Centre	Altnagelvin Hospital	1
215	Belfast	Downpatrick	Knockbracken Healthcare Park	1
215	Downpatrick	Belfast	Knockbracken Healthcare Park	1
3A	Derry-Londonderry City Centre	Lisnagelvin	Altnagelvin Hospital Glenshane Road	1

Live Data Reporting

```
//Executes a SQL reporting query dynamically to  
retrieve live data from the database. – Line 195  
const result = await runQuery(sql);
```

```
// Clears and rebuilds the report table dynamically  
without reloading the page. Line 171 + 246  
output.textContent = "";  
output.appendChild(table);
```

Aggregation

– Count Hospital Stops per Route

Counts hospital stops for each route
(different stops)

```
// Aggregates the number of hospital stops per route  
for reporting. – Line 177  
COUNT(DISTINCT s.stopId) AS hospitalStopCount;
```

5. REPORT – Accessibility Features by Stop Rosa_StopFeatures.html

Translink

Add Records Update Records Delete Records Complaint Reports Fea Rep

What accessibility-related features are available at each stop across the network?

This report allows operators to identify which stops currently provide accessibility aids and where additional infrastructure may be required.

Stop Search

Old Hollywood Road - Bus Stop

Confirm Stop Reset

Number of Features at Selected Stop: 1

Feature Name
Tactile Paving

This report highlights the accessibility infrastructure currently available at individual stops and helps identify locations where additional features may be required to improve inclusive access across the network.

No Rows Returned

Please select a valid route.

No accessibility features are currently recorded for the selected stop.

Parameterisation

The user selects a stop, which determines the specific accessibility features that are retrieved.

```
//Passes the selected stop ID as a parameter to the reporting query. - Line 261  
printTable(stopMap[stopText]);
```

```
//Restricts results to accessibility features for the selected stop. - Line 270  
WHERE sf.stopId = ${stopId}
```

Live Data Reporting

Data is being retrieved from the database rather than hard coded

```
// Clears and rebuilds the report table dynamically without reloading the page. - Line 267 + 305  
output.textContent = "";  
output.appendChild(table);
```

6. REPORT – Vehicle Accessibility Coverage

Rosa_WheelAccessiblePercentage.html



Use of SQL View

```
//Uses a database VIEW to classify vehicle  
accessibility before aggregation. – Line 196 + 274  
FROM vw_vehicle_accessibility
```

Aggregation – Fleet-Level Coverage

```
// Aggregates the number of vehicles in each  
accessibility category for reporting. – Line 193 - 195  
SELECT accessibilityStatus,  
COUNT(*) AS vehicleCount
```

Dynamic Chart Rendering

```
// Retrieves live data from the database and renders  
the chart dynamically without reloading the page. –  
Line 199 + 202  
const result = await runQuery(sql.trim());  
createDoughnutChart(result.data);
```

7. Data Retrieval – Dynamic and Parameterised Queries

The screenshot shows a web application interface with three main sections:

- Top Section:** A dropdown menu with the selected option "City Hospital Train Station - Train Station". Other visible options include "Tamnamore Park and Ride - Bus Stop", "Titanic Quarter Train Station - Train Station", "Sydenham Train Station - Train Station", and "Holywood Train Station - Train Station". Below this are two buttons: "Elevator Access" and "Hospital Stop".
- Route Search Section:** A search bar with the placeholder text "Search route by route name...". To its right is a "Confirm Route" button. Below the search bar, a list of routes is shown, with "1 - Belfast City to Bangor" and "1 - Bangor to Belfast" visible.
- Stop Search Section:** A search bar with the placeholder text "Search stop by stop name...". To its right is a "Confirm Stop" button. Below the search bar, a list of stops is shown, with "Bangor" selected. Other visible stops include "Bangor Academy", "Bangor Bus & Rail Centre", and "Bangor Donaghadee Road".

Below the Stop Search section, there is a "Stop Type" dropdown menu with the following options: "Train Station", "-- select an option --", "Bus Stop", "Train Station" (highlighted in blue), "Bus & Train Station", and "Buscentre".

Parameterised Query Pages

createStop.html, createRouteStop.html, updateStop.html, deleteFeature.html, Rosa_StopFeatures.html

Dynamically Updating Pages

createStop.html, createRouteStop.html, updateStop.html, deleteFeature.html, Rosa_Hospitals.html, Rosa_StopFeatures.html, Rosa_WheelAccessiblePercentage.html

Dynamic & Parameterised Data Retrieval

- User selection determines the query parameters
- SQL is executed dynamically
- Results update without any need for a page refresh

```
//Retrieves live database data dynamically based on user input without page reloads.
```

```
- Rosa_StopFeatures.html - Line 261
```

```
printTable(stopMap[stopText]);
```